

SWYAM VATWANI

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SUMMARY

Machine Learning enthusiast with practical experience in implementing and evaluating regression models from scratch, performing feature engineering, data preprocessing, and exploratory data analysis using Python. Experienced in developing AI-powered applications and applying machine learning techniques to real-world datasets. Passionate about building intelligent systems and continuously expanding expertise in **ML** and **Deep Learning**.

EDUCATION

Bachelor of Science in Computer Science • **CGPA: 9.0**

Aug 2024 - May 2028

National Institute of Technology, Delhi

PROJECTS

StudyGPT | AI-Powered Learning Platform

- Developed an AI-powered learning platform using **Retrieval-Augmented Generation (RAG)**, **MongoDB Atlas Vector Search**, **Gemini Embeddings**, **LLM-based reranking**, and a **multi-model AI pipeline**. 04/2026
- Implemented automated flashcard generation, quiz generation, summarization, and concept explanation features to enhance learning outcomes.
- Designed a retrieval pipeline combining **vector search**, embedding-based **semantic search**, and reranking to improve response relevance and accuracy.
- Integrated LLM-as-a-Judge evaluation to assess and validate AI-generated responses for quality and reliability.

Regression Models & Data Preprocessing from Scratch

06/2026

- Implemented Linear, Polynomial, and **Regularized Regression models** from scratch using **Python** and **NumPy**, including gradient descent optimization and model evaluation.
- Built end-to-end data preprocessing pipelines with **feature scaling**, **normalization**, **feature engineering**, train-validation splitting, and exploratory data analysis (EDA).
- Trained and evaluated models on real-world datasets from **Kaggle** using metrics such as MSE, RMSE, and R² Score, visualizing results with **Pandas** and **Matplotlib**.

WORK EXPERIENCE

Freelance Web Developer

French Language Learning Platform | Freelance | 2026

- Developed and deployed a production-ready language learning platform using the **MERN** stack, **Sanity CMS**, and **Mailchimp** for marketing automation. Configured and deployed the application on **Vercel** with custom domain integration through **GoDaddy**.
- Integrated **analytics and monitoring** tools to track user behavior and website performance.
- Achieved **230+** website visitors after launch.
- Enhanced search engine visibility through SEO best practices, sitemap generation, and optimized metadata.

ACHIEVEMENTS

- Supervised Machine Learning: Regression and Classification — DeepLearning.AI & Stanford University** ([Coursera](#)), 2026 
- Solved 300+ DSA problems across **LeetCode** and coding platforms.
- JEE ADVANCED, 2022** - Secured All India Rank of **4576** in general category (among 1,80,000 candidates).
- JEE MAINS, 2024** - Secured All India Rank of **5885** in general category (among 14,76,557 candidates).

SKILLS

MACHINE LEARNING: TENSORFLOW, SCIKIT-LEARN, NUMPY, PANDAS, MATPLOTLIB

PROGRAMMING: PYTHON, C++, JAVA

BACKEND: NODE.JS, EXPRESS.JS, FLASK

FRONTEND: REACT.JS, TAILWIND CSS

DATABASE: MYSQL, MONGODB ATLAS